

Job Aggregation Platform - Final Project Report (Manager Version)

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July 15, 2026

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Job Aggregation Platform — Final Project Report (Manager Version)

Internal Engineering Report · Data Platform

Prepared for	Engineering Manager
Prepared by	Reuben Jacob
Report date	July 15, 2026
Data snapshot as of	July 10, 2026
Companion document	<code>Final_Project_Report_v2.md</code> (full technical detail, methodology, and evidence trail)

This is a condensed version of `Final_Project_Report_v2.md`, written for a single reading in approximately 15–20 minutes. No technical finding has been removed — long narrative explanations have been replaced with tables, and repeated explanations have been cut. All figures are sourced from this project's PostgreSQL database, source code, and `job_api.xlsx` research workbook (API Catalog and Sheet1). Where a fact could not be verified from these sources, this report states **"Not publicly available"** or **"Not verified during this project"** rather than estimating.

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1. Project Overview

Purpose. Build a single, de-duplicated, queryable dataset of job postings, sourced from as many legitimately accessible providers as possible, normalized into one schema with country and remote/hybrid/on-site classification, and persisted in a real database rather than a flat file.

Scope. The project researched **268 job-related APIs, ATS platforms, and job boards** (government labour agencies, commercial aggregators, ATS platforms, and job boards), narrowed that list to an **86-provider shortlist** for close audit, and built **46 working integrations** against a shared `sources/` interface (`BaseJobSource`). Every researched provider — implemented or not — is documented with a specific, evidence-based reason (Section 4).

Final outcome.

- **46 providers implemented; 41 currently contributing live data** to PostgreSQL.
- **184,506 unique job records** stored, deduplicated by URL/hash, with per-provider run history.
- **Offline country and work-arrangement normalization** — no paid geocoding service used.
- The remaining **222 researched providers** were not implemented for specific, documented reasons: no public API, enterprise/partner-only access, wrong data-flow direction (employer-posting tools, not read APIs), deprecated services, or a registration/verification step not yet completed.
- The platform reflects **one complete backfill run**, not yet a recurring schedule — it is ready for operational hardening, not further architecture, before production use.

2. Project Status

Area	Status	Detail
Provider research (268 researched, 86 shortlisted)	Completed	Every provider carries a documented reason for its outcome (Section 4)
46 provider integrations (shared <code>BaseJobSource</code> interface)	Completed	One module per provider under <code>sources/</code>
PostgreSQL persistence (<code>jobs</code> , <code>provider_runs</code> tables)	Completed	Upsert-on- <code>dedup_key</code> , idempotent schema
Deduplication (URL/hash-based)	Completed	0 cross-provider duplicate URLs remaining
Offline country normalization	Completed	Multi-country/region text forced to <code>NULL</code> rather than guessed
Work-arrangement classification (remote/hybrid/on-site)	Completed	Derived from location text and provider-native flags
<code>jobs.json</code> static export + frontend search/filter	Completed	Unchanged since Milestone 2
Per-source fault isolation	Completed	One broken provider cannot halt a run
41 of 46 providers contributing live data	Partially Completed	5 non-contributing for external reasons (billing, missing credentials, one zero-result run) — Section 3
Employment-type capture	Partially Completed	Present for 24 of 46 providers; absent for the rest
Country resolution long tail	Partially Completed	Multi-country case solved; ~1,565 distinct unresolved single-place values remain
Credentialed-provider setup	Partially Completed	2 of 12 credentialed providers (Trade Me, CareerOneStop) lack credentials in this environment
6 real, accessible candidate providers (Saramin, VDAB, CBOP, Work24, GOV.UK Find a Job, Job-Room Switzerland)	Partially Completed	Registration/verification step outstanding, not rejected
Standardized job-category taxonomy	Not Started	No cross-provider taxonomy exists in the data (Section 5, item 11)
Salary / visa-sponsorship capture	Not Started	No schema field; no provider exposes it in a usable form
Scheduled/automated recurring syncs	Not Started	Every run to date has been a single manual invocation
Inline normalization at insert time	Not Started	Currently a separate, manually-triggered migration script
Provider-run health alerting	Not Started	No automated flag for a source dropping to 0 jobs
Dashboard/analytics UI	Not Started	Static search frontend only
Expired-job / posting-removal detection	Not Started	Requires a second ingestion run to diff against; none has occurred

3. Implemented APIs

3.1 LIFECYCLE SUMMARY

Stage	Count
Providers researched (API Catalog sheet, <code>job_api.xlsx</code>)	268
Providers shortlisted for close audit (Sheet1, <code>job_api.xlsx</code>)	86
Providers implemented (<code>sources/*.py</code> , registered in <code>sources/__init__.py</code>)	46
Providers currently contributing live data	41
Providers implemented but not currently contributing	5
Providers researched but not implemented	222

3.2 WHY 5 IMPLEMENTED PROVIDERS ARE NOT CURRENTLY LIVE

Provider	Reason
TheirStack	Account's billing plan does not include Jobs API access (HTTP 402)
Trade Me Jobs	Built, but no production credentials configured; also requires Trade Me's approval process
CareerOneStop	Built, but no credentials configured (<code>CAREERONESTOP_USER_ID</code> / <code>API_TOKEN</code>)
Fantastic.jobs	Credentialed, but returned 0 jobs in the latest run — most plausibly a trial-window/account-tier issue, not confirmed
Arbeitsnow	Free, no-auth API that previously worked (931 jobs in earlier testing); returned 0 jobs in the latest run — cause not confirmed

3.3 VOLUME CONCENTRATION

Provider	Jobs Stored	% of Total
SmartRecruiters	65,392	35.4%
Greenhouse	22,715	12.3%
Lever	17,980	9.7%
NHS Jobs	10,843	5.9%
USAJOBS	10,000	5.4%
Bundesagentur für Arbeit	9,896	5.4%
Reed	8,994	4.9%
NAV Arbejdsplassen	5,394	2.9%
Ashby	5,252	2.8%
Workable	4,137	2.2%

Top 3 providers supply 57.4% of all 184,506 stored jobs — volume is concentrated in ATS boards, not evenly spread across all 46 sources.

4. Complete Provider Research (268 Providers)

Every provider researched in this project (`job_api.xlsx` API Catalog sheet, 268 rows), split into **Implemented (46)** and **Not Implemented (222)**. Pricing model and payment terms are this project's own verified classification of each provider's published terms — where nothing could be confirmed, the cell reads "Not publicly available." Reference is a documented URL/domain found in this project's research notes.

4.1 IMPLEMENTED PROVIDERS (46)

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs (Stored / Fetched)	Status
SmartRecruiters	No-auth public API	No	developers.smartrecruiters.com/docs/posting-api	65,392 stored (65,394 fetched)	Live
Greenhouse	No-auth public API	No	developers.greenhouse.io/job-board.html	22,715 stored (22,715 fetched)	Live
Lever	No-auth public API	No	github.com/lever/postings-api	17,980 stored (17,980 fetched)	Live
NHS Jobs	No-auth public feed	No	jobs.nhs.uk/api/v1/search_xml	10,843 stored (13,011 fetched)	Live
USAJOBS	Free registration	No	developer.usajobs.gov	10,000 stored (10,000 fetched)	Live
Bundesagentur für Arbeit	No-auth-style public key	No	github.com/bundesAPI/jobsuche-api	9,896 stored (10,000 fetched)	Live
Reed	Free registration	No	reed.co.uk/developers/jobseeker	8,994 stored (9,000 fetched)	Live
NAV Arbeitsplassen	No-auth-style public token	No	navikt.github.io/pam-stilling-feed	5,394 stored (12,478 fetched)	Live
Ashby	No-auth public API	No	developers.ashbyhq.com/docs/public-job-posting-api	5,252 stored (5,252 fetched)	Live
Workable	No-auth public API	No	apply.workable.com (public widget surface)	4,137 stored (7,627 fetched)	Live
Workday	No-auth public API	No	(keyless endpoint, templated per tenant — no fixed docs URL found)	2,997 stored (2,998 fetched)	Live
Himalayas	No-auth public API	No	himalayas.app/jobs/api	2,253 stored (3,000 fetched)	Live

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Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs (Stored / Fetched)	Status
Arbetsförmedlingen	No-auth public API	No	jobsearch.api.jobtechdev.se	2,096 stored (2,100 fetched)	Live
Adzuna	Free tier + paid metered	Yes, beyond free tier/quota	developer.adzuna.com/docs/search	2,000 stored (2,000 fetched)	Live
The Muse	No-auth public tier	No	themuse.com/developers/api/v2	2,000 stored (2,000 fetched)	Live
CareerJet	Free registration	No	careerjet.com/partners/api	1,996 stored (2,000 fetched)	Live
EURES	No-auth public feed	No	europa.eu/eures (no official public docs)	1,681 stored (2,000 fetched)	Live
RemoteJobs.org	No-auth public API	No	remotejobs.org (in-response docs link 404s)	1,500 stored (1,500 fetched)	Live
France Travail	Free self-service	No	francetravail.io/produits-partages/catalogue/offres-emploi	1,146 stored (1,150 fetched)	Live
Jooble	Free registration	No	jooble.org/api	1,137 stored (1,137 fetched)	Live
Get on Board	No-auth public tier	No	getonbrd.com/api-doc.html	1,011 stored (1,261 fetched)	Live
Taiwan Ministry of Labor	No-auth open data	No	free.taiwanjobs.gov.tw (data.gov.tw dataset 44062)	1,000 stored (1,000 fetched)	Live
Findwork.dev	Free registration	No	findwork.dev/developers	1,000 stored (1,000 fetched)	Live
Mustakbil.com	No-auth public feed	No	rss.mustakbil.com/jobs-rss	500 stored (500 fetched)	Live
Teamtailor	No-auth public feed	No	docs.teamtailor.com	494 stored (494 fetched)	Live
Recruitee (Tellent)	No-auth public API	No	docs.recruitee.com/reference/intro-to-careers-site-api	394 stored (394 fetched)	Live

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Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs (Stored / Fetched)	Status
RemoteOK	No-auth public API	No	remoteok.com/api	100 stored (100 fetched)	Live
MyJobMag	No-auth public feed	No	myjobmag.com/feeds	100 stored (100 fetched)	Live
Jobicy	No-auth public API	No	jobi.cy/apidocs	100 stored (100 fetched)	Live
We Work Remotely	No-auth public feed	No	weworkremotely.com/remote-jobs.rss	99 stored (100 fetched)	Live
4dayweek.io	No-auth public feed	No	4dayweek.io/feed	50 stored (50 fetched)	Live
SerpApi (Google Jobs)	Free tier + paid metered	Yes, beyond free tier/quota	serpapi.com/google-jobs-api	50 stored (50 fetched)	Live
OpenWebNinja (JSearch)	Free tier + paid metered	Yes, beyond free tier/quota	openwebninja.com/api/jsearch	49 stored (49 fetched)	Live
Working Nomads	No-auth public API	No	workingnomads.com/jobs	33 stored (33 fetched)	Live
Remotive	No-auth public API	No	remotive.com/api-jobs	30 stored (30 fetched)	Live
Freshersworld	No-auth public feed	No	freshersworld.com/feed	30 stored (30 fetched)	Live
Jobspresso	No-auth public feed	No	jobspresso.co/jobs/feed	20 stored (20 fetched)	Live
MyCareersFuture	No-auth public API	No	api.mycareersfuture.gov.sg (no public docs found)	20 stored (2,000 fetched)	Needs review
NoDesk	No-auth public feed	No	nodesk.co/remote-jobs	10 stored (10 fetched)	Live
Hasjob	No-auth public feed	No	hasjob.co/feed	6 stored (6 fetched)	Live
HK Gov Vacancies	No-auth open data	No	csb.gov.hk/datagovhk/gov-vacancies (data.gov.hk)	1 stored (58 fetched)	Needs review
TheirStack	Per-credit metered (1 credit/job)	Yes	theirstack.com	0 stored (0 fetched)	Blocked — billing

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs (Stored / Fetched)	Status
Trade Me Jobs	Free (registered app)	No	developer.trademe.co.nz	0 stored (0 fetched)	Built, uncredentialed
CareerOneStop	Free registration	No	careeronestop.org/Developers/WebAPI/web-api.aspx	0 stored (0 fetched)	Built, uncredentialed
Fantastic.jobs	Free trial → paid metered (~\$1/1,000 jobs)	Yes, after trial ends	developer.fantastic.jobs	0 stored (0 fetched)	Credentialed, 0 result
Arbeitsnow	No-auth public API	No	arbeitsnow.com/api/job-board-api	0 stored (0 fetched)	0 in latest run

4.2 NOT IMPLEMENTED PROVIDERS (222)

Grouped by research classification at the time of this project. **Registration/Approval Required** and **Needs Live Verification** providers have a real, plausible access path (Section 6, Recommendations); **Not Viable** providers were rejected on structural grounds (no public API, enterprise-only, wrong data-flow direction, deprecated, or a dataset rather than a live API).

Registration/Approval Required (4)

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
CBOP - Centralna Baza Ofert Pracy (Poland public employment offices)	Free	No	praca.gov.pl	Potentially large - national public employment office coverage	Poland's national public-employment-office job database. Real SOAP web service returning batches of up to 1,000 live offers per file, but capped at 20 calls/cycle and only reachable during a 17:00-07:00 window -...
Saramin Open API (사람인 오픈 API)	Not publicly available (partner-gated access; pricing undisclosed)	Not publicly available	saramin.co.kr	Unknown - Saramin is one of Korea's largest commercial job boards but no public...	One of Korea's largest commercial job boards. Real documented API, but gated behind an application/approval step, explicitly labeled beta with a 500-calls/day cap - usable, just slower to unlock and rate-constrained.
VDAB Vacature API (Belgium - Flanders public employment service)	Free, but partner/approval access required (no self-serve signup)	No, once approved, but partner/approval access is itself the...	Not publicly available	Potentially large - Flanders regional coverage	Flanders' public employment service vacancy API - free but requires an application form and an exploratory onboarding meeting with VDAB before credentials are issued. Real read API, just a slower registration path than...
Work24 (formerly Worknet/워크넷) Job Listings Open API - Korea Employment Information Service	Free	No	Not publicly available	Potentially large - South Korea's national public employment portal (renamed 고용24 in...	South Korea's national public job bank, verified live on data.go.kr with real-time XML/JSON updates and a free key. Caveat: license text restricts use to non-commercial, attributed, unmodified use - confirm terms with...

Needs Live Verification (4)

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
GOV.UK Find a Job (DWP)	Free	No	Not publicly available	Potentially large	UK's national government job service; no confirmed public read/search API for third parties was found - only an employer-side SFTP bulk-upload path for posting vacancies. Worth one more direct check with DWP given its...
Job-Room Jobs API (SECO, Switzerland)	Free, but partner/approval access required (no self-serve signup)	No, once approved, but partner/approval access is itself the...	Not publicly available	Potentially large	Switzerland's biggest job platform, but the API is built primarily for employers/ATS submitting ads under a reporting-obligation rule - whether third-party aggregators can get read-only access is unconfirmed in the...
JobsPikr	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	Not publicly available	Unknown (volume not published)	Commercial aggregator (incl. Naukri/Monster-sourced India feeds) with a real API and docs, but pricing is custom/quote-based rather than a published self-serve rate - confirm actual signup process and cost before...
Portal del Empleo / Servicio Nacional de Empleo (STPS Mexico)	Free	No	Not publicly available	Potentially large	Mexico's national employment service publishes datasets on the CKAN-based datos.gob.mx open-data platform, but a live, current-vacancy-specific API endpoint for the job board itself was not directly confirmed - may...

Not Viable at Research Time (214)

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
104 Job Bank (104人力銀行)	Not applicable, no public API	Not applicable, no public API	104.com.tw	~1.1 million (110萬, Feb 2026)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
1111 Job Bank (1111人力銀行)	Not applicable, no public API	Not applicable, no public API	1111.com.tw	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
51job (前程无忧 / Qiancheng Wuyou)	Not applicable, no public API	Not applicable, no public API	51job.com	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
AMS eJob- Room (Austria public employment service)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
APEC (Association Pour l'Emploi des Cadres, France)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
APS Jobs (Australian Public Service careers)	Not applicable, no public API	Not applicable, no public API	apsjobs.gov.au	Moderate	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Adecco India	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Agencia Publica de Empleo SENA (APE)	Not applicable, no public API	Not applicable, no public API	edu.co	Potentially large	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
AllJobs.co.il	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	AllJobs.co.il	Potentially large	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Apideck	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	apideck.com/pricing	Not publicly available	Paid middlew re-wraps othe (several of w/ Greenhouse, Workday - we integrate dire free). Redunc enterprise-pri
Apify	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	apify.com/pricing	Not publicly available	Generic web-infrastructure not a job-spe using it mear and maintain custom per-s ourselves, w/ project in itse more provide
Apna	Not applicable, no public API	Not applicable, no public API	Not publicly available	Large, blue-collar (board size; no public access)	No public AP party aggreg;
Appcast	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	appcast.io/how-programmatic-recruitment-advertising-improves-cost-per-hire/	Not publicly available	Advertiser-sic distribution/C network - bui employers to postings out, aggregators l postings in. N read API.
Aura (by Aura Intelligence)	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Proprietary workforce database spanning 20M+ companies (billions of data points;...	Enterprise lal market-intelli; SaaS sold via contract - not serve job-pos feed.
BMET / Ami Probashi (Bangladesh Overseas Employment)	Not applicable, no public API	Not applicable, no public API	amiprobashi.com	Potentially large	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
BOSS Zhipin (Kanzhun Limited, NASDAQ: BZ)	Not applicable, no public API	Not applicable, no public API	zhipin.com	Not publicly available	No public AP site/portal rec jobseeker or login; no dev

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
					access is doc anywhere.
BambooHR	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available (N/A - per-tenant only, no aggregate feed)	Enterprise-or pricing/acces behind a sale partner-apprc process, no s developer pa
Bayt.com	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Bayt.com	Potentially large	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Bdjobs.com	Not applicable, no public API	Not applicable, no public API	Not publicly available	Potentially large	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Bolsa Nacional de Empleo (BNE / SENCE, Chile)	Free	No	Not publicly available	Potentially large	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Bossjob	Free	No	bossjob.com	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Bright Data	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	brightdata.com/products/web-scraper/jobs-scraper	116.8M+ records (general Jobs dataset); LinkedIn Jobs dataset specifically 62.4M+	Generic web-infrastructure not a job-spe using it mear and maintain custom per-s ourselves, w/ project in itse more provide
BrighterMonday	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	brightermonday.co.ke	Potentially large	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
Broadbean (Veritone Hire)	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	broadbean.com/product-suite/job-distribution-platform/	Not publicly available	Advertiser-side distribution/C network - built employers to post out, aggregators to post in. No read API.
Built In	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (tech/startup; not quantified)	No public API site/portal recruitment jobseeker or login; no dev access is documented anywhere.
Bumeran	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API site/portal recruitment jobseeker or login; no dev access is documented anywhere.
CEIPAL	Not publicly available	Not publicly available	ceipal.com/pricing	Not publicly available	No public API site/portal recruitment jobseeker or login; no dev access is documented anywhere.
CV-Library	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available	No public API site/portal recruitment jobseeker or login; no dev access is documented anywhere.
CWJobs	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	CWJobs.co.uk	12,000+ jobs (per cwjobs.co.uk About Us page)	No public API site/portal recruitment jobseeker or login; no dev access is documented anywhere.
CareerBuilder	Not publicly available (partner-gated access; pricing undisclosed)	Not publicly available	careerbuilder.com/	Not publicly available	No public API site/portal recruitment jobseeker or login; no dev access is documented anywhere.
CareerBuilder / Monster (Talent Network distribution)	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	careerbuilder.com/talent-network	Not publicly available	Advertiser-side distribution/C network - built employers to post out, aggregators to post in.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
					postings in. No read API.
CareerLink Vietnam	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
CareerOne (Australia)	Not publicly available (partner-gated access; pricing undisclosed)	Not publicly available	careerone.com.au	100,000+ (live jobs)	Feed-ingest / distribution n only - no pub endpoint for p data as a cor
Careers24	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	careers24.com	Potentially large	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Careers@Gov	Not applicable, no public API	Not applicable, no public API	Not publicly available	Small (SG public sector; not quantified)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Catho Developer API	Not publicly available (partner-gated access; pricing undisclosed)	Not publicly available	Not publicly available	Potentially large	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
China MOHRSS National Public Employment Service Platform (全国公共招聘服务平台)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Civil Service Jobs	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (not quantified)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Claro Analytics (WilsonHCG)	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	claroanalytics.com/	Real-time job-postings and 500M+ global talent profiles	Enterprise lal market-intelli; SaaS sold via contract - not serve job-pos feed.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
Cliclavoro / Ministero del Lavoro Open Data (Italy)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Computrabajo	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Coresignal	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	399M+ multi-source records (incl. India)	Enterprise-tie vendor sold v contact, not a published sel price - same the other enti data vendors
Cutshort	Not applicable, no public API	Not applicable, no public API	Not publicly available	Small (tech; board size not public)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Darwinbox	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available (N/A - per-tenant only, no aggregate feed)	Enterprise-or pricing/acces behind a sale partner-apprc process, no s developer pa
Datapeople (by Payscale)	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	datapeople.io/	Not publicly available (N/A (analyzes customer job postings, not a broad aggregation...	Enterprise lal market-intelli; SaaS sold via contract - not serve job-pos feed.
Dice	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (tech niche; not quantified)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Diffbot	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	diffbot.com/pricing	Not publicly available	Generic web-infrastructure not a job-spe using it mear and maintain custom per-s ourselves, w/

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
					project in its more provide
Draup	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	draup.com/	Global labor-market and job-postings data (aggregated multi-source)	Enterprise IaaS sold via contract - not serve job-post feed.
Drushim	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	Drushim.co.il	Potentially large	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Egypt Ministry of Labour (formerly Ministry of Manpower) Job Portal	Not applicable, no public API	Not applicable, no public API	labour.gov.eg	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Elemplo	Not applicable, no public API	Not applicable, no public API	Not publicly available	Very large (CO; not quantified)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Eluta	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Employment Services of South Africa (ESSA) - Dept. of Employment and Labour	Free	No	labour.gov.za	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Forasna	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	forasna.com/employer/pricing	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Foundit (Monster India)	Not applicable, no public API	Not applicable, no public API	Not publicly available	~800k+ (board size; no public access)	No public AP party aggreg;
Freshteam	Enterprise	Yes, enterprise/sales	Not publicly available	Not publicly available (N/A - per-tenant only,	Enterprise-or pricing/acces

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
		pricing, no self-serve free tier		no aggregate feed)	behind a sale partner-appro process, no s developer pa
Glassdoor Jobs API	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	glassdoor.com/developer/index.htm	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Glints	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Google Cloud Talent Solution (Job Search API)	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	google.com/talent-solution/pricing	Not publicly available (N/A (you populate it with your own jobs; not an aggregation...	This is a host matching/sea engine you p with your OW postings - it is source of agg third-party lis Enterprise infrastructure not a job fee
Google for Jobs	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	schema.org	Unknown (indexes open web)	Advertiser-sic distribution/C network - bui employers to postings out, aggregators 1 postings in. N read API.
Greenwich.HR (WageScape)	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	amazon.com/marketplace/pp/prodview-t5fvgfurgypxg	~5M new US jobs/month; ~70% of all open US jobs; 1.5M+ hiring organizations	Enterprise ial market-intelli; SaaS sold via contract - not serve job-pos feed.
GreytHR	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available (N/A - HRMS/payroll only, no job feed)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
GulfTalent	Not applicable to data access (fee is for employers posting jobs on	Not applicable to data access (see Pricing Model column)	gulftalent.com	Potentially large	No public AP site/portal rec jobseeker or login; no dev

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
	this platform, not...				access is doc anywhere.
Gupy Public API	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	gupy.io	~90,000 job vacancies published monthly on the Gupy R&S platform (official figure)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
HackerEarth Jobs	Not applicable, no public API	Not applicable, no public API	Not publicly available	Small (board size not public)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
HasData	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	hasdata.com/prices	Not publicly available	Generic web-infrastructure not a job-spe using it mean and maintain custom per-s ourselves, w/ project in itse more provide
Hello Work Job Information API (ハローワーク求人情報提供サービス, MHLW)	Free	No	Not publicly available	Potentially large - nationwide public employment service covering hundreds of thousands...	Japan's real, documented job API - but registration is to licensed pi placement bu local governr training institi generic aggre does not qua
HelloWork ATS Partner API (France)	Free, but partner/approval access required (no self-serve signup)	No, once approved, but partner/approval access is itself the...	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Hired.com (LHH Recruitment Solutions)	Not applicable, no public API	Not applicable, no public API	Hired.com	0	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Hirist	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (tech; board size not public)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
Hong Kong Labour Department Interactive Employment Service (iES, jobs.gov.hk)	Not applicable, no public API	Not applicable, no public API	www.jobs.gov.hk	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Horsefly Analytics	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	horseflyanalytics.com/	1 trillion+ data points; millions of job postings; supply/demand across 60+ countries	Enterprise lal market-intelli; SaaS sold via contract - not serve job-pos feed.
IEFPOne / netEmprego (Portugal public employment service)	Not applicable, no public API	Not applicable, no public API	Not publicly available	5,913 ofertas de emprego (live count)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Indeed Hiring Lab	Free	No	indeed.com	Not publicly available (N/A (publishes indices/aggregates, e.g. Job Postings Index, Wage...	Indeed's ecoi research arm publishes ag; labor-market indices/trend (mirrored on not individual records. Research/sta product, not i listings feed.
Indeed India	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Huge (not accessible - partner-only)	Enterprise pa access; not a a generic ag
Indeed PLUS Job Distribution API (Recruit Holdings, Japan)	Not publicly available (partner-gated access; pricing undisclosed)	Not publicly available	Not publicly available	Potentially large - aggregates postings across Japan's major job boards distributed via...	Advertiser-sic distribution/C network - bui employers to postings out, aggregators i postings in. N read API.
InfoJobs (Spain)	Not applicable, no public API	Not applicable, no public API	Not publicly available	68,383 ofertas (live count, Spain, as of July 2026)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Instahyre	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (tech; board size not public)	No public AP site/portal rec jobseeker or

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
					login; no dev access is doc anywhere.
Internshala	Not applicable, no public API	Not applicable, no public API	Not publicly available	Large (intern/fresher; board size not public)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Israeli Employment Service (Sherut HaTaasuka / taasuka.gov.il)	Free	No	taasuka.gov.il	Potentially large	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
JSearch (RapidAPI, by OpenWeb Ninja)	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	rapidapi.com/letscape-6bRBa3QguO5/api/jsearch	Not publicly available	Generic B2B aggregation \ a dedicated j paid, and ove sources alrea integrated.
Jadarat (Saudi National Employment Portal / HRDF)	Free	No	Not publicly available	Potentially large	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Job Bank (ESDC) feed / Open Gov dataset	Free	No	Not publicly available	Very large (100,000+)	Canada's nat bank: the live requires ESD approval, and fallback open a monthly CS missing employer/cor names and p apply URLs - fields this pro schema...
Job Market Finland / Tyomarkkinatori (TE-palvelut, Finland)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
JobKorea	Not applicable, no public API	Not applicable, no public API	Not publicly available	~209,637 (live listings)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
JobStreet / JobsDB (SEEK group)	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Very large if accessible (not public)	SEEK-group boards for SE enterprise pa access, no se path (catalog marks 'Not Recommend
JobThai	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	jobthai.com	~121,000 active jobs (121,337 shown live on jobthai.com)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Jobberman	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	jobberman.com	Potentially large	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Jobcase (MaxRecruit)	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	jobcase.com/products/maxrecruit	Not publicly available	Advertiser-sic distribution/C network - bui employers to postings out, aggregators l postings in. N read API.
Jobcase Platform API	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	jobcase.com/	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Jobg8	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	jobg8.com/Faq.aspx?contentid=10874	Not publicly available	Advertiser-sic distribution/C network - bui employers to postings out, aggregators l postings in. N read API.
Jobillico	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Jobindex (Denmark)	Not applicable, no public API	Not applicable, no public API	Not publicly available	33,600 jobs (live) / 30,000+ job ads	No public AP site/portal rec jobseeker or

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
					login; no dev access is doc anywhere.
JobisJob (LIFULL Connect)	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	jobboardfinder.com/jobboard-jobisjob-usa	~9,000,000 (monthly, cross-country)	Advertiser-side distribution/C network - built employers to post out, aggregators to post in. No read API.
Jobnet.dk (Denmark public employment portal)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API site/portal rec jobseeker or login; no dev access is doc anywhere.
Jobrapido	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	jobrapido.com	Not publicly available	No public API site/portal rec jobseeker or login; no dev access is doc anywhere.
Jobs.cz	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Medium (CZ/SK; not quantified)	Enterprise-or pricing/access behind a sale partner-app process, no self-developer pa
JobsDB Hong Kong (SEEK Group) / CTgoodjobs	Not applicable, no public API	Not applicable, no public API	hk.jobs	~32,700 (JobsDB HK) / ~32,000 (CTgoodjobs)	No public API site/portal rec jobseeker or login; no dev access is doc anywhere.
JobsEQ (Chmura Economics & Analytics)	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	chmura.com/jobseq	Real-time job-posting dataset with international coverage	Enterprise-level market-intelligence; SaaS sold via contract - not self-serve job-post feed.
JobsIreland.ie (Ireland public employment service)	Not applicable, no public API	Not applicable, no public API	Not publicly available	4,806 vacancies (live count)	No public API site/portal rec jobseeker or login; no dev access is doc anywhere.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
Jobsite	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Jobsite.co.uk	280,000+ live job adverts (per jobsite.co.uk About Us page)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
JoinVision (JobCloud HR Tech)	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	crunchbase.com/organization/joinvision	Not publicly available	Advertiser-sic distribution/C network - bui employers to postings out, aggregators l postings in. N read API.
Jora (Australia)	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	jora.com	Potentially large	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Joveo	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	joveo.com/programmatic-job-advertising-platform/	Not publicly available	Advertiser-sic distribution/C network - bui employers to postings out, aggregators l postings in. N read API.
JustRemote	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	justremote.co/remote-jobs	2,000+ hidden remote jobs	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Kalibr	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Karir.com	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Keka	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available (N/A - HRMS/payroll only, no job feed)	No public AP site/portal rec jobseeker or login; no dev

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
					access is doc anywhere.
Kelly Services India	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Kemnaker Karirhub (SIAPkerja)	Free, but partner/approval access required (no self-serve signup)	No, once approved, but partner/approval access is itself the...	Not publicly available	Potentially large	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Kombo.dev	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	kombo.dev/pricing	Not publicly available	Paid middlew re-wraps othe (several of w/ Greenhouse, Workday - we integrate dire free). Redunc enterprise-pri
Ladders	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (niche; not quantified)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Lensa	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	lensa.com/	Not publicly available	Advertiser-sic distribution/C network - bui employers to postings out, aggregators l postings in. N read API.
Lightcast (formerly Emsi Burning Glass) Job Postings API	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	lightcast.dev/apis/job-postings	Billions of postings aggregated from 160,000+ sources (18B+ labor-market data points)	Enterprise lal market-intelli; SaaS sold via contract - not serve job-pos feed.
Lightcast Jobs Canada	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	Not publicly available	Not publicly available	Paid labor-m: intelligence p sold through process, not serve develop signup; catal marks 'Not Recommend

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
LinkUp	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	linkup.com	Millions of postings indexed daily from 80,000+ employer websites	Enterprise lal market-intelli; SaaS sold via contract - not serve job-pos feed.
LinkedIn Jobs	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Huge (not accessible - partner-only)	Enterprise pa program only serve develop for job search/aggre
MOHRE UAE Federal Careers / iRecruitment Portal	Not applicable, no public API	Not applicable, no public API	mohre.gov.ae	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
MP Rojgar	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (state-level; not quantified)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
MahaJobs	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (state-level; not quantified)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
ManpowerGroup India	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Mantiks	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	mantiks.io/pricing	Not publicly available	Generic B2B aggregation \ a dedicated j paid, and ove sources alrea integrated.
Merge.dev	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	merge.dev/pricing/unified	Not publicly available	Paid middlew re-wraps oth (several of w Greenhouse, Workday - we integrate dire free). Redunc enterprise-pri

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
Mitula (LIFULL Connect / Adzuna)	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	lifullconnect.com/brands/mitula/	Not publicly available	Advertiser-side distribution/C network - built employers to post out, aggregators to post in. No read API.
Monster	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	monster.com/solutions/talent-management/partner-with-monster/	Not publicly available	No public API site/portal recruitment jobseeker or login; no dev access is documented anywhere.
NAVTTTC National Employment Exchange Tool (NEXT / jobs.gov.pk)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Potentially large	No public API site/portal recruitment jobseeker or login; no dev access is documented anywhere.
NEAIMS - National Employment Authority Kenya	Free	No	Not publicly available	Not publicly available	No public API site/portal recruitment jobseeker or login; no dev access is documented anywhere.
National Career Service (NCS)	Free	No	data.gov.in	Large in aggregate	India's NCS is a periodic bulk government (data.gov.in), per-job search stale snapshot real-time position Doesn't fit a large architecture.
National Directorate of Employment (NDE) Nigeria	Not applicable, no public API	Not applicable, no public API	nde.gov.ng	Not publicly available	No public API site/portal recruitment jobseeker or login; no dev access is documented anywhere.
National Job Portal (NJP) - Pakistan	Not applicable, no public API	Not applicable, no public API	njp.gov.pk	Small-to-moderate	No public API site/portal recruitment jobseeker or login; no dev access is documented anywhere.
National Job Portal -	Not applicable, no public API	Not applicable, no public API	jobs.gov.bd	Not publicly available	No public API site/portal recruitment jobseeker or

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
Bangladesh (jobs.gov.bd)					login; no dev access is doc anywhere.
Naukri	Not applicable, no public API	Not applicable, no public API	Not publicly available	~1M+ (board size; no public access)	No public AP access is undocumente area and Indi restricted.
Naukrigulf	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	naukrigulf.com	Potentially large	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Nexxt (formerly Beyond.com)	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	Beyond.com	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Nimble (Nimbleway)	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	nimbleway.com/pricing	Not publicly available	Generic web-infrastructure not a job-spe using it mear and maintain custom per-s ourselves, wl project in itse more provide
OCC Mundial	Not applicable, no public API	Not applicable, no public API	Not publicly available	137,925 active vacancies ("vacantes activas en occ.com.mx", as of Oct 18, 2024);...	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Outsourcely	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	outsourcely.com	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Oxylabs	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	oxylabs.io/products/scraper-api/web/pricing	Not publicly available	Generic web-infrastructure not a job-spe using it mear and maintain custom per-s ourselves, wl project in itse more provide

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
PandoLogic (pandoIQ) / Veritone Hire	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	pandologic.com/solutions/programmatic-advertising-pandoiq/	Not publicly available	Advertiser-side distribution/C network - built employers to post out, aggregators to post in. No read API.
Pangian	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	pangian.com/feed/?post_type=job_listing	Varies (curated remote listings)	pangian.com (301) to a static You for Being Pangian' shun notice on GitHub Pages (pangianhq.github.io) confirmed live other path (/jobs/about/, /blog candidate UF
People Data Labs	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	peopledatalabs.com/docs/job-posting-data-overview	Not publicly available	Generic B2B aggregation via a dedicated job paid, and over sources already integrated.
PhilJobNet (DOLE Bureau of Local Employment)	Free	No	philjobnet.gov.ph	Potentially large	No public API site/portal recruitment jobseeker or login; no developer access is documented anywhere.
Piloterr	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	piloterr.com/pricing	Not publicly available	Generic web-infrastructure not a job-specific using it means and maintain custom per-se ourselves, with project in itself more provide
Pnet	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	pnet.co.za	Potentially large	No public API site/portal recruitment jobseeker or login; no developer access is documented anywhere.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
Portal Empleo Argentina (Ministerio de Capital Humano / ex-Trabajo)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Potentially large	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Pracuj.pl (Poland)	Not applicable, no public API	Not applicable, no public API	Not publicly available	762,827 job offers published in 2025	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
PredictLeads	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	predictleads.com/pricing	270M+ records since 2018; ~9.8M active jobs	Generic B2B aggregation \ a dedicated j paid, and ove sources alre integrated.
Profesia	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Medium (CZ/SK; not quantified)	Enterprise-or pricing/acces behind a sale partner-appr process, no s developer pa
Public Service Commission of Sri Lanka (psc.gov.lk)	Not applicable, no public API	Not applicable, no public API	psc.gov.lk	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Quess Corp	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Radancy	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	radancy.com/en/programmatic-adtech/	Not publicly available	Advertiser-sic distribution/C network - bui employers to postings out, aggregators l postings in. N read API.
Randstad India	Not applicable, no public API	Not applicable, no public API	Not publicly available	1,394 jobs (live count on official site)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
Recruit CRM	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available (N/A - per-tenant only, no aggregate feed)	Enterprise-or pricing/access behind a sale partner-app process, no self-serve developer pa
Recruitics	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	recruitics.com/programmatic-job-advertising	Not publicly available	Advertiser-side distribution/C network - built employers to post out, aggregators to post in. No read API.
Remote First Jobs (Dynamite Jobs)	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	remotefirstjobs.com/rss	Varies (curated remote-first listings)	remotefirstjobs.com behind an Amazon Cloudflare challenge on path tested (i.e. robots.txt it successfully confirmed live 403 with a 'Just a moment...' in challenge page).
Remote.co	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	remote.co/feed	Not publicly available	No public API site/portal recruitment jobseeker or login; no developer access is documented anywhere.
Revelio Labs	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	reveliolabs.com	1.1B+ workforce profiles plus job postings, transitions, sentiment, layoffs	Enterprise-level market-intelligence; SaaS sold via contract - not self-serve job-posting feed.
Rikunabi / Mynavi (Recruit Co. / Mynavi Corporation)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public API site/portal recruitment jobseeker or login; no developer access is documented anywhere.
Rojgar Sangam (UP)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (state-level; not quantified)	No public API site/portal recruitment jobseeker or login; no developer access is documented anywhere.
Rozee.pk	Not applicable, no public API	Not applicable, no public API	Not publicly available	Potentially large	No public API site/portal recruitment

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
					jobseeker or login; no dev access is documented anywhere.
SEEK (SEEK.com.au / SEEK.co.nz)	Free, but partner/approval access required (no self-serve signup)	No, once approved, but partner/approval access is itself the...	SEEK.com.au	Potentially large	Feed-ingest / distribution not only - no public endpoint for job data as a cor
SEPE Open Data Portal (Spain public employment service)	Free	No	Not publicly available	Not publicly available	Spain's SEPE publishes statistical/annual open-data file confirmed live API for current vacancies excluding known third-party scraper for it deprecated.
SINE Aberto (Sistema Nacional de Emprego / Ministerio do Trabalho e Emprego)	Free	No	dados.gov.br/dataset/sine-aberto	Potentially large	Brazil's SUSPER data-sharing October 2022 (CODEFAT R 956/2022) per LGPD privacy compliance until no active API integrate again now.
ScraperAPI	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	scraperapi.com/pricing	Not publicly available	Generic web-infrastructure not a job-specific using it means and maintain custom per-site ourselves, with project in itself more provide
Scrapingdog	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	scrapingdog.com/pricing	Not publicly available	Generic web-infrastructure not a job-specific using it means and maintain custom per-site ourselves, with project in itself more provide
Shine	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (board size; no public access)	No public API site/portal record jobseeker or login; no dev

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
					access is doc anywhere.
SmartDreamers	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	smartdreamers.com/pricing	Not publicly available	Advertiser-side distribution/C network - built employers to post out, aggregators to post in. No read API.
Snagajob	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	snagajob.com/	Not publicly available	No public API site/portal for jobseeker or login; no dev access is doc anywhere.
StepStone	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	stepstone.com	Not publicly available	No public API site/portal for jobseeker or login; no dev access is doc anywhere.
Superset	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Medium (campus; per-institution, not aggregatable)	Enterprise-level pricing/access behind a sales partner-application process, no self-developer pay
Talent.com	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Talent.com	40 million+ jobs (global figure stated on talent.com homepage)	No public API site/portal for jobseeker or login; no dev access is doc anywhere.
TalentNeuron (Gartner)	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	talentneuron.com/solutions/platform	Real-time labor data covering ~90% of world GDP; 40TB normalized data, 3B+ profiles,...	Enterprise-level market-intelligence; SaaS sold via contract - not self-serve job-posting feed.
Talroo	Not applicable to data access (fee is for employers posting jobs on this platform, not...)	Not applicable to data access (see Pricing Model column)	talroo.com/pro/	Millions (per Talroo publisher marketing)	Advertiser-side distribution/C network - built employers to post out, aggregators to post in. No read API.

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
TeamLease	Not applicable, no public API	Not applicable, no public API	Not publicly available	23,082+ active job vacancies (live count on official site)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Techmap.io	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	jobdatafeeds.com	292M+ available jobs	Generic B2B aggregation v a dedicated j paid, and ove sources alrea integrated.
Telangana T-Jobs	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (state-level; not quantified)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Textkernel Jobfeed / Market IQ	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	textkernel.com/products-solutions/labour-market-insights/market-iq/	Large multi-country vacancy corpus (millions of postings, deduplicated/enriched)	Enterprise lal market-intelli; SaaS sold via contract - not serve job-pos feed.
Thailand Department of Employment - Smart Job Center (ไทยมีงานทำ)	Free	No	Not publicly available	Potentially large	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
The Burning Glass Institute	Free	No	burningglassinstitute.org/	Not publicly available (N/A (research using Lightcast and other data))	Enterprise lal market-intelli; SaaS sold via contract - not serve job-pos feed.
TimesJobs	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (board size; no public access)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
TopCV	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	Not publicly available	Potentially large	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
TotalJobs	Enterprise	Yes, enterprise/sales	Totaljobs.com	Not publicly available	No public AP site/portal rec

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
		pricing, no self-serve free tier			jobseeker or login; no dev access is doc anywhere.
Triplebyte	Free, but partner/approval access required (no self-serve signup)	No, once approved, but partner/approval access is itself the...	triplebyte.com/	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Trovit (LIFULL Connect / Adzuna)	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	trovit.com/feed-in-jobs/	Not publicly available	Advertiser-sic distribution/C network - bui employers to postings out, aggregators 1 postings in. N read API.
UWV Open Match Data (Netherlands)	Free	No	Not publicly available	Not publicly available (N/A - aggregated statistics only, not individual postings)	UWV (Dutch employment publishes ag labor-market datasets, not live job postir per-job identi in the data at
Unified.to	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	Not publicly available	Not publicly available	Paid middlew re-wraps othe (several of w Greenhouse, Workday - we integrate dire free). Redunc enterprise-pri
Unstop	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (campus/intern; not a job-listing API)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Upward.net (a Jobcase company)	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	upward.net/employers	Not publicly available	Advertiser-sic distribution/C network - bui employers to postings out, aggregators 1 postings in. N read API.
Vagas for Business API (Vagas.com.br)	Not publicly available (partner-gated	Not publicly available	Vagas.com	Not publicly available	No public AP site/portal rec jobseeker or

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
	access; pricing undisclosed)				login; no dev access is doc anywhere.
VietnamWorks Open API	Not publicly available (partner-gated access; pricing undisclosed)	Not publicly available	vietnamworks.com	Potentially large	Partner Access Required - go approved cor partners, not developer req
WUZZUF	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	wuzzuf.net	Potentially large	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Welcome to the Jungle	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Wellfound	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (startup roles; board size not public)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
WhatJobs	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	whatjobs.com/job-aggregators	Not publicly available	Advertiser-side distribution/C network - bui employers to postings out, aggregators 1 postings in. N read API.
Work and Income / Find a Job (MSD, New Zealand)	Not applicable, no public API	Not applicable, no public API	workandincome.gov	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Work at a Startup (YC Combinator)	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	workatastartup.com	Unknown (thousands across 1,000+ YC companies)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
WorkIndia	Not applicable, no public API	Not applicable, no public API	Not publicly available	Large, blue-collar (board size; no public access)	No public AP site/portal rec jobseeker or login; no dev

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
					access is doc anywhere.
Workato	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	workato.com/pricing	Not publicly available	Paid middlew re-wraps othe (several of w/ Greenhouse, Workday - we integrate dire free). Redunc enterprise-pri
Workforce Australia	Free	No	workforceaustralia.gov.au	Potentially large	The public Af write/manage (employers c manage job : OAuth2 throu Business por public job-list read/search : exists to pull from.
Workopolis	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
WowJobs	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
Xing Jobs	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
XpressJobs (xpress.jobs)	Not applicable, no public API	Not applicable, no public API	Not publicly available	Not publicly available	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
ZipAlerts (ZipRecruiter TrafficBoost)	Not applicable to data access (fee is for employers posting jobs on this platform, not...	Not applicable to data access (see Pricing Model column)	zipalerts.com/	Not publicly available	Advertiser-sic distribution/C network - bui employers to postings out, aggregators i

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
					postings in. No read API.
ZipRecruiter API	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Large (not quantified)	Enterprise-or pricing/access behind a sale partner-apprec process, no self-developer pay
Zoho Recruit	Enterprise	Yes, enterprise/sales pricing, no self-serve free tier	Not publicly available	Not publicly available (N/A - per-tenant only, no aggregate feed)	Enterprise-or pricing/access behind a sale partner-apprec process, no self-developer pay
ZonaJobs	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	zonajobs.com.ar	Not publicly available	No public API site/portal recruitment jobseeker or login; no developer access is documented anywhere.
Zyte (formerly Scrapinghub)	Free Tier + Pay-as-you-go	Yes, beyond the free tier/quota	zyte.com/zyte-api/pricing.html	Not publicly available	Generic web-infrastructure not a job-specific using it means and maintain custom per-selves, with project in itself more provide
data.gov.in OGD platform	Free	No	Not publicly available	Varies (per-resource)	General-purpose open-government portal spanning sectors; not job-specific, and resources are per-resource rather than a postings feed
eQuest	Paid (metered/flat; see Cost Details column source for exact terms)	Yes, paid from the outset, no free tier documented	equest.com/	Not publicly available	Advertiser-side distribution/C network - built employers to postings out, aggregators for postings in. No read API.
iimjobs	Not applicable, no public API	Not applicable, no public API	Not publicly available	Medium (management roles; board size not public)	No public API site/portal recruitment jobseeker or login; no developer

Provider	Pricing Model	Payment After Free Tier	Official Reference	Approx. Jobs Available	Reason Not Implemented
					access is doc anywhere.
jobs.ch (Switzerland)	Not applicable, no public API	Not applicable, no public API	Not publicly available	45,825 jobs (live)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
karriere.at (Austria)	Not applicable, no public API	Not applicable, no public API	github.com/karriereat	13,400+ jobs (live)	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.
topjobs.lk	Not applicable, no public API	Not applicable, no public API	Not publicly available	Potentially large	No public AP site/portal rec jobseeker or login; no dev access is doc anywhere.

5. Internship Requirement Coverage

2. FINAL PROJECT REPORT

This document, together with the companion technical report `Final_Project_Report_v2.md`, is the final project report.

3. APIS/SERVICES USED

46 providers implemented (Section 3); 268 researched in total (Section 4).

4. COMPLETED VS. PENDING

See the [Project Status table](#) (Section 2).

5. FULLY / PARTIALLY / NOT STARTED

See the [Project Status table](#) (Section 2).

6. PRICING MODEL (FREE / FREE TRIAL / PAY-AS-YOU-GO / SUBSCRIPTION / ENTERPRISE)

Pricing Model (Implemented Providers)	Count	Providers
Free (no cost, ever)	41	40 fully free/no-auth or free-registration providers, plus Trade Me Jobs (free, but approval-gated)
Free Trial (time-limited, then paid)	1	Fantastic.jobs (7-day trial, then metered)
Pay-as-you-go / Metered (has a paid, usage-based component)	5	Adzuna, SerpApi, OpenWebNinja, TheirStack, Fantastic.jobs
Subscription (flat recurring fee)	0	Not used by any implemented provider
Enterprise (sales-negotiated contract)	0	Not used by any implemented provider — 15 enterprise-only candidates were identified in research and not implemented (Section 4.2)

Full per-provider pricing model is in Section 4.1. The 222 not-implemented providers' pricing (where verifiable) is in Section 4.2.

7. PAYMENT REQUIRED AFTER FREE TIER

Answer	Count (of 46 implemented)
No (free with no paid tier)	41
Yes, beyond a free tier/quota, or after a trial	5

8. PRICING / DOCUMENTATION LINKS

All 46 implemented providers have an official reference URL, quoted directly from source code (Section 4.1). Of the 222 not-implemented providers, a documented reference (domain or URL) was found in this project's research for **166 of 268 providers overall**; for the remainder, the reference cell reads "Not publicly available" rather than a guessed link (Section 4.2).

9. NUMBER OF JOBS

Metric	Value
Total unique jobs stored (PostgreSQL)	184,506
Total jobs fetched, all providers, pre-dedup	200,727
Distinct companies represented	18,367

Top providers by volume are in Section 3.3. Approximate job counts for not-implemented providers (where publicly known) are in Section 4.2.

10. REGION COVERAGE

Region	Dedicated Providers	Notes
United States	4	USAJOBS, CareerOneStop, Findwork.dev, Adzuna
Canada	0	No dedicated source; incidental coverage only via global ATS boards
United Kingdom	2	Reed, NHS Jobs
Europe (non-UK)	5	Bundesagentur für Arbeit, Arbeitsförmedlingen, France Travail, EURES, NAV Arbeidsplassen
India	2	Freshersworld, Hasjob
LATAM	1	Get on Board
APAC	4	MyCareersFuture, Taiwan MOL, HK Gov Vacancies, Trade Me
Africa	1	MyJobMag
Middle East	0	No single-country dedicated source (Mustakbil spans 20+ countries, counted as global)
Global / Remote / Multi-country	27	All 6 ATS platforms plus 21 aggregator/remote-board providers

Resulting job distribution is US-heavy (47.3%), then UK (14.3%) and Germany (6.9%) — a sourcing-mix effect from which countries have dense curated ATS company lists, not a normalization defect. Canada remains the clearest unresolved regional gap.

11. JOB CATEGORIES

No standardized, cross-provider job category exists in this dataset, and this cannot be supported without inventing one.

Verified directly against all 46 providers' `normalize()` implementations:

- **0 of 46** providers expose a job-category field that means the same thing across providers.
- **23 of 46** expose a provider-specific department, team, business-unit, occupation, or function field (e.g. SmartRecruiters' `function`, Taiwan Ministry of Labor's occupation code, USAJOBS's federal job series) — each proprietary to that one provider.
- **23 of 46** expose no category-like field at all (confirmed in code, e.g. Reed's module comment: "no job-category/employment-type field at all").

Evidence that these fields cannot be merged: the stored tag "Engineering" comes from SmartRecruiters' `function` field (6,797 jobs), Greenhouse's `departments` field (1,154 jobs), and Ashby's `department` / `team` field (1,136 jobs) — three structurally unrelated fields that happen to use the same English word. Treating them as one taxonomy would silently merge unrelated data. This project's schema stores all of this in a single free-text `tags` array; a real crosswalk would require manually mapping each provider's raw values, which has not been done.

12. DATA QUALITY

Dimension	Finding
Missing fields	Confirmed per-provider in code — e.g. Reed exposes no employment-type field at all; no provider exposes usable salary data
Company info	43 of 46 providers give a direct company-name field; 3 derive it (Ashby, Lever, Workday)
Duplicate rate	8.08% overall (see item 14 below)
Stale postings	11.9% of stored jobs are over 1 year old

13. JOB FRESHNESS

Metric	Value
Jobs with a posted date recorded	184,129 of 184,506 (99.8%)
Median posting age	22 days
Mean posting age	229 days (skewed by a long tail of old postings on some boards)
Posted within 30 days of snapshot	108,003 (58.6%)
Older than 1 year	21,915 (11.9%)

Refresh frequency and expired-job detection are **not verified during this project** — the dataset reflects one ingestion run, not a recurring schedule, so there is no interval or diff to measure yet.

14. DUPLICATE ANALYSIS

Metric	Value
Total in-run duplicates skipped (all providers)	16,221
Total fetched, pre-dedup	200,727
Overall deduplication rate	8.08%
Cross-provider duplicate URLs remaining	0

Highest duplicate rates: MyCareersFuture (99.0%, `limit` param silently ignored by the provider), HK Gov Vacancies (98.3%, every job shares one generic portal URL), NAV Arbejdsplassen (56.8%, a change-feed rather than a current-postings search), Workable (45.8%, the same job posted to multiple cities, correctly collapsed).

15. RATE LIMITS

Limit type	Examples
Hard result-window ceiling	USAJOBS and Bundesagentur für Arbeit (10,000 records); Reed (~9,900, returns HTTP 500 beyond it)
Monthly quota	SerpApi and OpenWebNinja (200 requests/month free tier)
Per-job billing	TheirStack (1 credit per job returned)

The examples above are the limits discovered and documented during implementation (`Final_Project_Report_v2.md`, Section 7); not all 46 providers' rate limits were independently re-verified for this report.

16. RESPONSE TIME & RELIABILITY

Provider	Duration (s)	Reliability
SmartRecruiters	385.7	High
Lever	870.1	High (slowest run despite modest volume — per-company pagination overhead)
Greenhouse	183.8	High
NHS Jobs	102.4	Undocumented endpoint (no official API contract)
USAJOBS	53.1	High
Bundesagentur für Arbeit	174.3	High (with retry logic)
Reed	56.6	High
NAV Arbeitsplassen	51.4	Rotating token (no fixed key)
Ashby	70.3	High
Workable	51.8	High

Full duration/reliability figures for all 46 providers exist in the underlying `provider_runs` table; only the ten highest-volume providers are shown here for brevity.

17. AUTHENTICATION

Auth Type	Count (of 46)
None (no auth required)	31
API key (registered)	10
Public non-secret key/token (no registration)	2
OAuth	2
HTTP Basic	1

18. FILTERING CAPABILITIES

Provider	Filtering Available
SmartRecruiters	Company, offset/limit
Greenhouse	Company only
Lever	Company, offset/limit
NHS Jobs	Keyword
USAJOBS	Page/ResultsPerPage only (no keyword/filter params implemented)
Bundesagentur für Arbeit	Keyword, location
Reed	Pagination only (no keyword/filter params implemented)
NAV Arbeitsplassen	Modified-since cursor
Ashby	Company only
Workable	Company only

Filtering capability was verified for the ten highest-volume providers; it was **not independently documented for the other 36 implemented providers** in this report. No provider was found to expose salary, experience-level, or visa-sponsorship filtering.

19. MUST HAVE / SHOULD HAVE / NICE TO HAVE

Priority	Item
Must Have	Move normalization inline into the save path
Must Have	Scheduled, automated recurring syncs
Must Have	Resolve the 5 non-contributing providers (credentials, billing, root-cause)
Should Have	Reporting dashboard on top of PostgreSQL
Should Have	Better location enrichment (~1,565 distinct unresolved country values)
Should Have	Normalized employment-type field; a department/occupation crosswalk table
Should Have	Dedicated Canada, broader APAC, and Middle East sources
Should Have	Provider-run health alerting
Should Have	Complete the 6 deferred candidate providers (Saramin, VDAB, CBOP, Work24, GOV.UK Find a Job, Job-Room Switzerland)
Nice to Have	Funding-stage/company-size enrichment
Nice to Have	Salary capture where a source exposes it
Nice to Have	Historical trend tracking across multiple runs
Nice to Have	Periodic re-verification of the 268-provider research catalog

20. PRODUCTION RECOMMENDATIONS

Summarized here; full detail in Section 6. Before production reliance: close the 5-provider contribution gap, add provider-run health alerting, move normalization inline, and put the pipeline on a recurring schedule.

6. Final Recommendations

Current readiness. The platform is functionally complete for a first release: 46 providers integrated, 184,506 deduplicated jobs stored in PostgreSQL, offline country/work-arrangement normalization applied, and full fault isolation so one broken provider cannot take down a run. It reflects **one complete backfill run**, not an ongoing sync.

Remaining work, in priority order:

1. Provision missing credentials (Trade Me, CareerOneStop), resolve TheirStack's billing block, verify Fantastic.jobs' trial/account status, and re-run Arbeitnow to confirm its 0-job result was transient.
2. Pursue the two highest-value deferred candidate providers — Saramin and VDAB — both real APIs blocked only by a registration/onboarding step.
3. Add provider-run health alerting before scheduling recurring syncs, so a source silently dropping to 0 doesn't go unnoticed.
4. Move country/work-arrangement normalization inline into the ingestion path, replacing the current manual migration script.
5. Investigate the two low-yield sources (HK Gov Vacancies, MyCareersFuture) before relying on either for APAC coverage.

Production readiness. Not yet production-scheduled. Recommended before relying on this pipeline for ongoing reporting: (a) resolve the 5-provider gap above, (b) add health alerting, (c) put the pipeline on a recurring cron/schedule, (d) confirm secrets handling carries through to the deployment target, (e) extend automated test coverage from country normalization to the full aggregation/dedup pipeline.

Next steps. Treat the items above as a scoped, well-understood backlog rather than unknowns — none of them block the current dataset from being useful today. The full technical detail, evidence, and provider-by-provider reasoning behind every recommendation is in `Final_Project_Report_v2.md`.

Job Aggregation Platform — Final Project Report (Manager Version) · Data snapshot 2026-07-10 · Report date July 15, 2026 · Condensed from `Final_Project_Report_v2.md` for a single 15–20 minute engineering-manager review. No technical finding was removed; long narrative sections were replaced with tables, and repeated explanations were cut. Source of truth: PostgreSQL (`jobs`, `provider_runs`), this repository's source code, and `job_api.xlsx` (API Catalog, Sheet1 sheets). Where a fact could not be verified, this report states “Not publicly available” or “Not verified during this project” rather than estimating.